Determination of Leaf Area Index (LAI) from hemispherical canopy Photographs

TASK 1: Calculate LAI by yourself

Introduction: Leaf Area Index (LAI) serves as a crucial parameter in understanding biosphere-atmosphere interactions. While remote sensing techniques are commonly employed for large-scale LAI estimation, field-based measurements are indispensable for calibration and validation purposes. Optical sensors, such as digital hemispherical photography, provide a feasible means to estimate LAI in the field. However, it's essential to acknowledge that field-based LAI measurements, termed effective LAI, rely on certain assumptions regarding canopy characteristics.

Objective: The objective of this report is to calculate LAI using the Beer-Lambert law of radiation attenuation from the given data.

Methodology: LAI is estimated by integrating transmissions (gap fractions) over zenith angles from 0 to π / 2 using the formula:

$$LAI = -2 \int_{0}^{\pi/2} \log(T(\theta)) \sin(\theta) \cos(\theta) \, d\theta$$

Which is approximated by the sum

$$LAI = -2 \sum_{i=1}^{n} \log(T_i) \cos(\theta_i) \, w_i$$

Data: Given the following data:

Ring	T	θ	w	Radian (r)	LN(T) * COS(r) * w
1	0.424	10.7	0.0473	0.18675031	-0.039878782
2	0.323	23.3	0.1025	0.40666189	-0.106388744
3	0.248	38.2	0.1577	0.66671606	-0.172798374
4	0.1	52.9	0.2034	0.92327957	-0.282509932
5	0.013	67.8	0.4891	1.18333374	-0.802558911
				SUM	**-1.404134744**
				LAI	**2.808269488**

Where, T = estimated gap fraction, ɵ = mean zenith angle, r = radian (ɵ/180) w = weight (fraction of hemisphere covered by the ring)

Results: Using the provided data, the LAI is **2.8082**

Conclusion: In this report, LAI was successfully calculated using the provided data and the Beer-Lambert law of radiation attenuation. LAI serves as a valuable parameter in understanding canopy structure and its interaction with incoming radiation, which is essential for various ecological and environmental studies. This method provides an efficient means to estimate LAI in the field, enabling researchers to validate and calibrate remote sensing techniques for broader-scale applications.

Task 2. Analyze a set of eight hemispherical photographs

S.N.	Image ID	Threshold value	Algorithms	LAI
1	DSCN5774	163	(auto, R. & C. (1978))	2.04
2	DSCN6013	-	detection separately for each ring, R. & C. (1978)	1.01
3	DSCN6108	163	(auto, R. & C. (1978))	1.53
4	DSCN6167	-	detection separately for each ring, R. & C. (1978)	1.78
5	DSCN6246	70	Manually	1.12
6	DSCN6330	110	Manually	0.09
7	DSCN6429	110	Manually	2.64
8	DSCN6462	96	Manually	1.26

1. What are ideal conditions for obtaining images, and what kind of images should be avoided?

The ideal conditions for obtaining images are:

Even Lighting: Images should be captured under uniform lighting conditions to ensure consistent brightness across the canopy. Uneven lighting can introduce bias in LAI estimates by affecting the perceived gap fractions.

Clear Sky: Clear skies without significant cloud cover minimize variations in sky brightness, leading to more accurate classification of canopy and sky pixels. Overcast or cloudy conditions can result in inconsistent classification, impacting LAI estimates.

Minimal Obstructions: The camera should have an unobstructed view of the canopy without obstructions such as branches, leaves, or other objects blocking the lens. Obstructions can distort the canopy-sky boundary, affecting the accuracy of gap fraction calculations.

Stable Camera Position: The camera should be securely mounted or held steady to prevent movement during image capture. Any movement can introduce blur or distortion, affecting the accuracy of pixel classification and subsequent LAI estimation.

The images to avoid are:

Poor Lighting Conditions: Images taken under low light conditions or with harsh shadows can result in inaccurate classification of canopy and sky pixels, leading to unreliable LAI estimates.

Cloudy or Overcast Weather: Images captured during cloudy or overcast weather can lead to inconsistent sky brightness, making it challenging to accurately distinguish between canopy and sky pixels.

Obstructed Views: Images with obstructed views, such as those containing significant amounts of foliage in the foreground or objects blocking the canopy, should be avoided as they can distort the canopy-sky boundary and affect pixel classification.

2. Which reasons could cause random variation or systematic bias in the optical LAI estimates?

Canopy Heterogeneity: Variations in canopy structure, such as differences in leaf density, canopy height, or species composition, can introduce random variation in optical LAI estimates. Heterogeneous canopies may exhibit varying degrees of light penetration, leading to inconsistencies in gap fraction calculations.

Shadow Effects: Shadows cast by trees, branches, or other structures within the canopy can create systematic bias in optical LAI estimates by affecting the perceived gap fractions. Shadows may obscure portions of the canopy, leading to underestimation of LAI in shadowed areas.

Lens Distortion: Optical distortion caused by the camera lens or fisheye converter can introduce systematic bias in LAI estimates by altering the apparent size or shape of canopy elements. Lens distortion may result in inaccuracies in pixel classification and gap fraction calculations.

Incorrect Thresholding: Inaccurate thresholding methods or settings can lead to both random variation and systematic bias in optical LAI estimates. Choosing an inappropriate thresholding method or setting can result in misclassification of canopy and sky pixels, affecting the accuracy of LAI calculations. Adjusting thresholding parameters based on image characteristics and lighting conditions is crucial to minimize errors.

Image Quality: Variations in image quality, such as resolution, focus, or noise levels, can introduce random variation in LAI estimates. Poor-quality images may contain artifacts or inconsistencies that impact pixel classification and LAI calculations, leading to unreliable results. Ensuring high-quality image capture and processing techniques is essential for minimizing random variation in optical LAI estimates.